

# FUNDAMENTALS OF COSTING AND PRICING

## DAY 1



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## LESSON FOUR

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COSTING IN THE AGRICULTURE SECTOR

COSTING BASICS – SEASONAL FINANCE: INTEREST

COSTING BASICS: CALCULATING THE PRICE OF  
ONE BROILER CHICKEN

## COSTING IN THE AGRICULTURE SECTOR

Costing in Agriculture follows similar rules to any other sector, but it does come with a couple of differences, which makes it an interesting case study. Below we discuss four main ones:

1. Length of the production cycle can extend across many months, or even years
  2. Relationship between inputs and outputs is not fixed, and is influenced by factors out of the farmer's control (eg temperature, day length, rainfall)
  3. High capital investment costs for land and equipment require repayment over an extended period
  4. Fixed pricing and market monopolies limit pricing options for certain produce.
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1. **Lengthy Production Cycles.** Certain crops, for example Macadamia nuts, cattle and oranges only begin bearing fruit many months and even years into the cycle. The longer the gap between the cost of starting the farm (planting the trees or buying the cows) the greater the need for capital to cover cashflow deficit. Careful planning and budgeting is needed to ensure the farm doesn't run out of money before the harvest happens and income starts to flow into the business.
  2. **Variable Input: output relationship.** Unlike simple manufacturing costing where one bracelet always uses 5 beads, the outputs in farming (e.g. chicken weight, time to harvest and yield) are determined not only by the inputs (fertiliser, seeds, water, food) but by other factors out of the farmer's control. This is one of the reasons why farming is such a tricky industry to be in, and the reason why farmers invest in irrigation, lighting and greenhouses – to try and control these other variables. The lack of a fixed input: output relationship also means that whilst costing can be forecast using a historical budget (ie. what results you got in the past) it also needs to be costed on a real-time basis - to check your actual results and price each time and with every harvest.
  3. **High Capital Cost.** Farming is notorious for its high capital cost – especially if you need to purchase land at commercial prices. In addition, the cost of buildings, irrigation equipment and tractors adds a huge price to the production cost. It is important that these are included in the costing models – but divided over a realistic period. Land for example is usually repaid over 25 years, tractors and buildings over 10 years and other equipment over a shorter period. Even if your land and equipment has been paid for already, your costing must build in the price of replacing this equipment, or you will have no funds to replace the buildings and machinery when they become old.
  4. **Fixed pricing** in agriculture is a challenge that often requires the farmer to reduce their cost of production in order to make a profit. The dairy industry is notorious for this worldwide, with the market price for milk fixed below breakeven in many first world countries – requiring farmers to exist on subsidies. Locally this impacts the small farmer who is dictated to by the market, which is much bigger and stronger than he/she is. Strategies for better pricing include producing 'out of season' when fewer people are selling and prices are higher, doing value-adds such as packaging and processing, and producing special, quality or unusual products – for example black tomatoes, green cauliflower and red runner beans.

## COSTING BASICS – SEASONAL FINANCE: INTEREST

Most commercial farmers have to present a production budget to the bank every year with a request for seasonal finance. This budget includes an estimate of the inputs (seed, fertiliser, electricity etc) and the projected yield and price showing income.

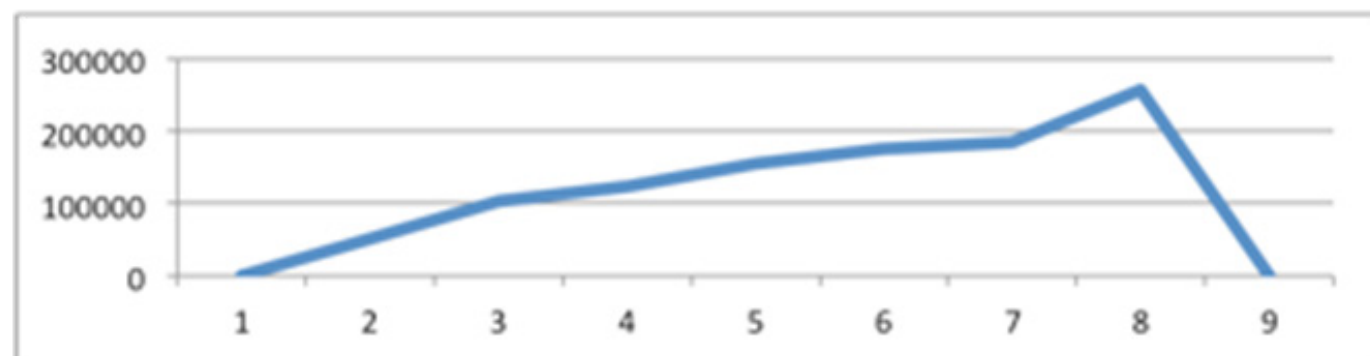
The bank manager needs to be convinced that the farmer is a 'good risk' and there is a good chance that he will repay the loan once the harvest has been made.

Seasonal finance will be given to the farmer to purchase the inputs, pay labour, pay electricity and all other costs for the entire growing season, and interest is calculated on the total overdraft on a monthly basis.

These interest costs need to be included in your costing, to ensure that you make a profit at the end. Challenges arise when a crop sale is delayed or when yields and/or prices are lower than expected, as bank managers generally just want their money on time!

### Sample: Simple Budget forecast for field crop – 9 month growing cycle

| Month     | 1      | 2           | 3         | 4        | 5         | 6        | 7        | 8        | 9       |
|-----------|--------|-------------|-----------|----------|-----------|----------|----------|----------|---------|
| Activity  | plough | Plant       | harrow    |          |           |          |          | Harvest  | Sell    |
| Activity  |        | Irrigate    | Irrigate  | Irrigate | Irrigate  | Irrigate |          |          |         |
| Activity  |        | Weed killer | pesticide |          | pesticide |          |          |          |         |
| Overdraft | 0      | R50,000     | R100,000  | R120,000 | R150,000  | R170,000 | R180,000 | R250,000 | R0      |
| Interest  | 0      | R833        | R1,667    | R2,000   | R2,500    | R2,833   | R3,000   | R4,167   | R17,000 |
| Total     | 0      | R50,833     | R102,500  | R123,667 | R154,500  | R175,333 | R185,833 | R257,167 | R0      |



## COSTING BASICS: CALCULATING THE PRICE OF ONE BROILER CHICKEN

Many farmers are really great at recording their total expenses, but struggle to work out how much one cabbage, one chicken or a kilogram of beans actually costs to produce.

The fundamentals are exactly the same as all costing models – namely that one must understand how to calculate the rating (how much is used in the production) and price per single unit (kg, litre) of input.

In broiler production, if we know how much feed etc goes into the chicken house, and we know how many chickens are in the house, it is possible to work out the cost per chicken quite easily.

## Let's look at an example:

Mr Ndlovu has 5 chicken houses on his farm, each housing 3000 chickens and taking an average 6 weeks to grow, plus a 2 week cleaning period – making an 8 week (2 month) production cycle in all.

His assumptions are

| ASSUMPTIONS                                 |      |
|---------------------------------------------|------|
| Day Old Chicks                              |      |
| 6 weeks from day-old to sale (8 week cycle) |      |
| Mortality rate                              | 10%  |
| Quantity of chicks per house                | 3000 |
| Number of houses on the farm                | 5    |
| Lifetime of chicken house (years)           | 7    |

### 1. Variable costs

Mr Ndlovu's direct (also called variable or operating costs) costs include feed, medicine and labour, which can be added up to get a total cost of R67,640 per house of 3000 chickens, as shown below:

|                                  | Qty      | Price per unit |          | Total per house | Comment                            |
|----------------------------------|----------|----------------|----------|-----------------|------------------------------------|
| Direct Costs (per house)         |          |                |          |                 |                                    |
| Day old Chicks                   | 3000     | 6.55           | Each     | 19,650          | Chicks per house                   |
| Starter Mash                     | 30       | 275.00         | Bags     | 8,250           | per 3000 chicks for 2 weeks        |
| Grower Feed                      | 60       | 264.00         | Bags     | 15,840          | per 3000 chicks for 2 weeks        |
| Finisher Feed                    | 70       | 265.00         | Bags     | 18,550          | per 3000 chicks for 2 weeks        |
| Shavings                         | 100      | 16.00          | Bags     | 1600            | allowance per house                |
| Medication (Vaccine)             | 1        | 250.00         | House    | 250.00          | allowance per house                |
| Labour                           | 2        | 1500.00        | Labourer | 3,000.00        | 2 individuals                      |
| Casual labour                    | 1        | 500.00         | Labourer | 500.00          | Allowance                          |
| Total Direct/<br>Operating Costs | R 67,640 |                |          |                 | Per chicken house per 8-week cycle |

## 2. Overhead or Fixed costs

We now need to work out what the overhead costs are, by adding the expenses per month for the farm as a total. To work out the farm overheads, simply add the monthly costs (phone, travel, insurance etc). In this example the total overheads for the farm, per month are R23,300.

| Overheads (per Farm) per month        |               |                             |
|---------------------------------------|---------------|-----------------------------|
| Electricity                           | 3600          | heaters, lighting per month |
| Cellphone                             | 1200          | Per farm per month          |
| Travel/fuel                           | 3500          | Per farm per month          |
| Director/s salary                     | 12000         | Per farm per month          |
| Repairs                               | 500           | Per farm per month          |
| Rates & taxes                         | 750           | Per farm per month          |
| banking & bookkeeper                  | 250           | Per month                   |
| Insurance                             | 1500          | Per farm per month          |
| <b>Total Farm Overheads per month</b> | <b>23,300</b> | <b>per month</b>            |

To get the cost for **one house** we divide this total by 5 (the number of houses) ( $23,300 / 5 = \text{R } 4660$ )

## 3. Capital Replacement costs

If the farmer already owns the land and buildings, his capital costs only include the cost of replacing the houses and equipment at the end of their useful lifetime (estimated at 7 years). [nb, if he were still paying off a loan the calculation includes interest and capital repayments. Find this example in the extra notes]

We calculate how much he must put aside (save) each month to ensure there are funds to replace the old buildings and equipment - by dividing the total capital cost over 7 years.

| Capital cost              |                  | Notes                                                     |
|---------------------------|------------------|-----------------------------------------------------------|
| Building & fittings       | 151791           | 3000 birds 30 x 7 m excludes concrete floor               |
| Extras                    | 1500             | Allowance for Power and electrical boxes and installation |
| Sundry 5%                 | 7664.55          | Other                                                     |
| <b>Total capital cost</b> | <b>R 160,956</b> | Per chicken house                                         |
| Depreciation/month        | R1,916           | 7 Years monthly over 7 years                              |

This replacement cost (also called **depreciation** in the accounts) is included as part of the overheads, to ensure that all the costs of production is accounted for. Note: this money should be saved as cash reserves so that you can replace things as they fall apart!

By adding the monthly depreciation/replacement cost of R1916 to the overhead costs we get a total monthly overhead cost per house **per month** ( $\text{R}4660 + \text{R}1916 = \text{R}6576$ )

#### 4. Overheads per chicken house, per cycle

We know that each batch of chickens takes 8 weeks or two months – so to calculate the total overheads per batch of 3000 chickens we multiply the total monthly overhead by 2 to get R13,152. ( $R6576 \times 2 = R13,152$ )

|                                             |                 |             |                          |
|---------------------------------------------|-----------------|-------------|--------------------------|
| Total Farm Overheads per month              | 23,300          |             | per month                |
| Monthly Overheads per house                 | 4,660           | 5 Houses    | Per month per house      |
| Depreciation/ replacement cost per house    | 1,916           | 1 per house | Per month per house*     |
| Overheads + depreciation per house          | 6,576           |             |                          |
| <b>Total Overheads per cycle (2 months)</b> | <b>R 13,152</b> |             | per (8 weeks) two months |

#### 5. Total direct costs plus overhead cost per chicken

- To work out the Total Cost, we add the Direct and Overhead costs for the house ( $R67,640 + 13152 = R80,792$ )
- To calculate cost per chicken we divide our Total Cost by the number of chickens sold (2700). \* Remember to factor in the mortality rate (losses) – in this case we started with 3000 chicks and estimated a 10% loss rate ( $3000 - 300 = 2700$ )

The total cost per chicken is therefore total costs divided by number sold = ( $R80,792/2700 = R30$ )

This gives us a cost per chicken, factoring in all our costs (direct, overhead etc) of R30 each

|                                  |          |   |                                    |
|----------------------------------|----------|---|------------------------------------|
| Total costs (Direct + Overheads) | R 80,792 |   | Per chicken house per 8-week cycle |
| Chickens sold                    | 2,700    | A | 10% mortality (1000 - 100)         |
| Cost per chicken                 | R 29.92  | B |                                    |

#### 6. Profit margin

Knowing that we sell our chickens at R40 each, we can work out our profit per chicken ( $R40$  minus  $R29.92 = R10.08$  profit), and our total income per chicken house ( $R10.08 \times 2700 = R27,207$ ).

| Revenue/Income             |            |   |
|----------------------------|------------|---|
| Selling Price per Chicken  | R40        | C |
| Profit per chicken (C-B+D) | R10.08     | D |
| Profit Per house (D x A=E) | R27,207.73 | E |

An interactive excel sheet is included in your training pack for you to use for your own business.

